

assets – money and bonds. Money is useful for transactions, but it yields no returns, while bonds offer interest income and possibility of making capital gains when bond prices rise. There is an inverse relation between the price of bonds and rate of interest. Now consider the effect of change in interest rate on speculative demand for money. When interest rate is high, people will mostly expect it to fall and expect bond prices to rise. Therefore, they would reduce money holdings and buy bonds in the expectation of making a capital gain. So, at high rates of interest, demand for money would be low. Conversely, at low rates of interest, people expect interest rates to rise (i.e., fall in bond prices). Capital loss would induce people to sell bonds and increase the demand for money. Therefore, speculative demand for money is *inversely* related to the rate of interest.

In general, the higher the market rate of interest, the higher is the opportunity cost (interest income forgone) of holding money. So when interest rates rise, people would reduce money demand and try to hold interest-bearing assets instead which explains the inverse relation between interest rates and the demand for money. This is another way of understanding the inverse relation between money demand and interest rates, without assuming the presence of speculative demand.

From our discussion above, you should see that the money demand function can be expressed as follows:

$$L = qY - t.i \quad \dots (2.15)$$

where $q > 0$, $t > 0$; L is the demand for real balances, Y is aggregate income and i is the rate of interest. Diagrammatically the money demand function can be shown as a downward sloping line in Fig. 2.10, where interest rate is presented along the vertical axis and money demand is along the horizontal axis.

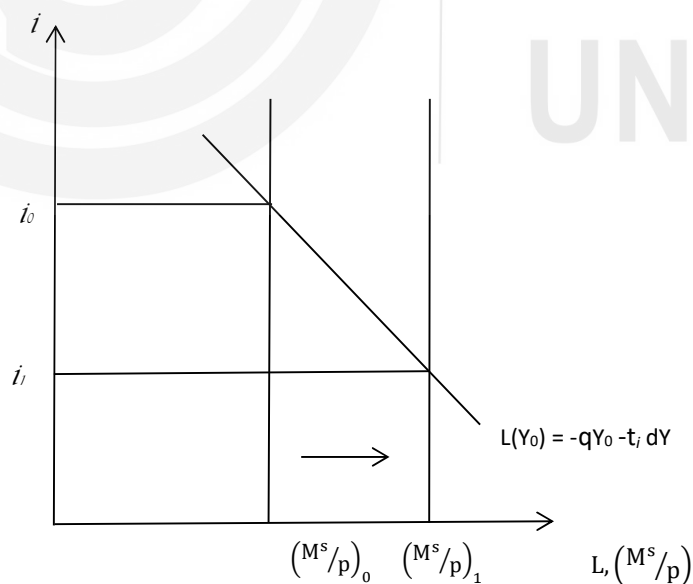


Fig. 2.10: Demand for Money

Note that the money demand function $L(Y_0)$ represents demand for money at the income level Y_0 . Movement along $L(Y_0)$ demonstrates the interest sensitivity of the money demand function. As interest rate falls from i_0 to i_1 , there is an increase in the demand for money due to speculative demand for money, as discussed above. For all income levels, demand for money greater than Y_0 would be higher at each interest rate (since transactions and precautionary demands for money increase with income), and the money demand curve would shift to the right. For incomes lower than Y_0 , money demand would be lower at each interest rate and the money demand curve would shift to the left.

Money supply is determined by the central bank of the country (you will learn more about it in subsequent Units) and is given by the vertical line M^s/P , where M^s is nominal money supply, P is the price level and M^s/P is the real money supply. Equilibrium in the money market occurs at the intersection of money supply and money demand. Note that with the downward sloping money demand function, an increase in money supply (from $(M^s/P)_0$ to $(M^s/P)_1$) brings about a reduction in the rate of interest (from i_0 to i_1).

2.5.2 Liquidity Trap

A special situation can arise if interest rates become unusually low and hit the lowest possible level, so that the demand for money becomes infinitely elastic and the economy faces a 'liquidity trap'. This can happen when interest rates have fallen so low (and bond prices have risen so high) that everyone expects interest rates to rise (and bond prices to fall). Therefore no one wants to hold bonds, in fear of making a capital loss and everyone wants to hold money. In this situation, the money demand curve becomes horizontal as presented in Fig. 2.11 and t in equation (2.16) tends to infinity.

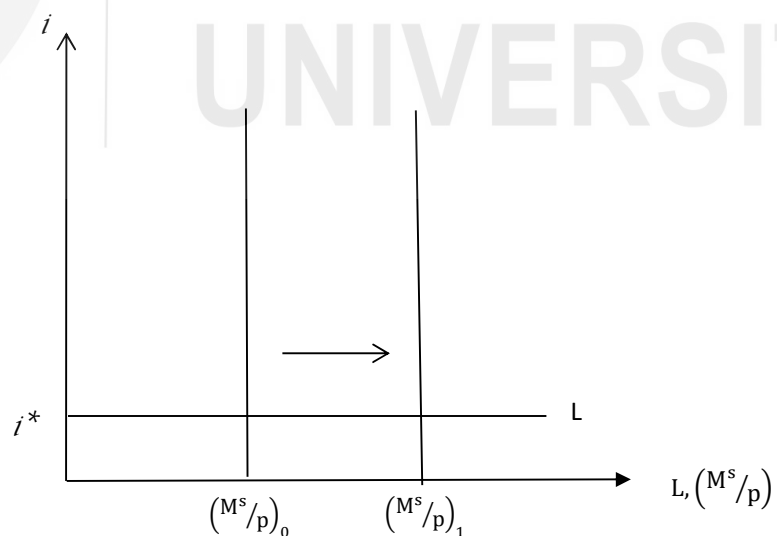


Fig. 2.11: Liquidity Trap

When interest rates hit a floor such as i^* , any injection of extra liquidity in the system is absorbed, as people willingly hold money (liquidity) rather than bonds at an unchanged rate of interest, i^* . Therefore once the economy is in a liquidity trap monetary policy cannot lower the rate of interest beyond i^* and hence it cannot affect real variables like s . An increase in money supply (from $(M^s/P)_0$ to $(M^s/P)_1$) has no impact on the interest rate which remains unchanged at i^* . Once you learn about monetary and fiscal policies, you will see that conventional monetary policy becomes ineffective when the economy is in a liquidity trap whereas fiscal policy is very effective in this case.

The concept of liquidity trap became important in policy circles during the global financial crisis of 2007-08, when in advanced economies like USA and Japan interest rates hit a floor and became zero (USA) or almost zero (Japan). In this case conventional monetary policy could not be used to reduce the rate of interest on government bonds any further and policy experiments involving the use of unconventional monetary policy measures had to be explored. Fiscal policy measures were also employed, but increase in public debt levels raised concerns regarding debt sustainability. In the next section we will explore the role of fiscal policy in the Keynesian model and touch on some of these policy issues.

2.6 ROLE OF GOVERNMENT

The classical economists, as we saw in Unit 1, advocated minimal government intervention in determination of prices and wage rate. According to them the activity of the government should be limited to law and order, justice and defence. They assumed that prices and wage rate are flexible, and there is perfect competition in the markets. Consequently, there is no unemployment in the economy.

The Keynesian economics is in sharp contrast to the classical economics. Keynesian model implies that the government has an important role to play in the economy. The government should play an active role to bring in equilibrium in the market. Keynes highlighted that rigidities in prices and wage rate prevent smooth functioning of market mechanism in the real world. He also highlighted the critical role of effective demand in determining the level of output and employment in a macroeconomic context. The classical economists had focused almost exclusively on the supply side (i.e., on production of output). In contrast, Keynesian analysis showed that aggregate demand (i.e., planned spending by economic agents) plays a key role in determining the level of output and employment in the short run.

Keynes pointed out that when lack of effective demand is the underlying cause of an economic recession. When there is high unemployment, there is tendency for wage rate to decline. This will worsen the situation by reducing purchasing power and hence effective demand. Therefore, government should intervene by

increasing public expenditure; it should go for a deficit budget. Fiscal and monetary policies can be used to inject additional demand in the economy, so as to enhance spending (aggregate demand). That would lead to increases in output, income and employment in the economy, and the economy would emerge out of the recession.

The success of Keynesian policies during the Great Depression (1929-33) made a strong case for an activist role of the government. In particular, Keynesian economists advocated the use of discretionary fiscal policies, as Keynes had pointed out the limitations of monetary policy. If the economy was in a liquidity trap, conventional monetary policy would be completely ineffective, while fiscal policy would still remain effective.

During the global financial crisis of 2007-08, most countries used the Keynesian prescription of expansionary fiscal policy (increases in government spending) to boost economic activity, although it resulted in widening budget deficits and high levels of public debt. High interest burden on public debt contributes to further widening of budget deficits, raising concerns about sustainability of the deficits. Further, high interest rate may lead to 'crowding out' of private investment.

You must be aware that during 2020 and 2021, the world economy suffered a severe economic crisis due to Covid-19. There was acute shortage in aggregate demand due to decrease in economic activity. The response of the governments to this crisis situation brought to the fore the importance of the policies advocated by Keynes.

Check Your Progress 4

1. Discuss some of the factors that influence the demand for money.

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2. What is speculative demand for money? How does it affect the slope of the money demand function?

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3. What is meant by 'liquidity trap'? What are its implications?

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4. Discuss the role of the government in an economy that is facing a prolonged recession.

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5. Give any three reasons why policy makers tend to be concerned about mounting budget deficits.

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2.7 LET US SUM UP

Keynesian model is appropriate for an economy facing recessionary conditions with unemployed resources and lack of effective demand. Keynes assumes that markets are imperfect with wage and price rigidity. Within this framework, output is demand-determined. An increase in any autonomous component of aggregate demand has a multiplier effect, leading to several rounds of increment in output. Our discussion on the consumption and investment functions highlighted the factors that influence these two important components of aggregate demand.

Equilibrium in the Keynesian model can be viewed in terms of the equality between planned saving and planned investment. We learnt about the paradox of thrift, which shows that an increase in the propensity to save results in a situation where aggregate saving may remain unchanged or decline. Since output is demand-determined in this model, an increase in planned saving leads to lower aggregate demand and hence lower output, explaining why we observe the paradox of thrift.

The demand for money (liquidity preference) increases with increase in income and decreases with increase in interest rate. The inverse relationship between

money demand and the rate of interest could be explained by the Keynesian concept of speculative demand for money. We also learnt about the liquidity trap – at unusually low rate of interest the demand for money becomes infinitely elastic and traditional monetary policy became ineffective.

2.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Consumption function is a linear function given by $C = a + bY$. The APC declines with increase in Y while the MPC remains constant.
- 2) i) A decrease in autonomous consumption will lead to a parallel downward shift in the consumption function.

(ii) An increase in MPC will increase the slope of the consumption function.
- 3) Autonomous investment remains constant irrespective of the level of income or interest. Induced investment depends on income or rate of interest or both. See Sub-Section 2.2.2 for details.
- 4) As interest rate decreases the cost of borrowing decreases. Thus, some of the projects which were unviable earlier become profitable. Therefore, more investment takes place with decrease in interest rate.

Check Your Progress 2

- 1) Keynesian model is based on the assumption that resources are under-utilised, prices and wage rate are rigid, and output is a homogenous commodity.
- 2) a) Aggregate demand function is: $Y = 30 + 0.75Y$. Equilibrium income is 120.

b) Equilibrium output will increase to 140.

c) 4

d) In the case of lump sum tax, the multiplier will be the same as autonomous expenditure multiplier given in (b). In the case of proportional tax at the rate of 0.1, the multiplier will be reduced to $\frac{1}{1-0.75 \times 0.9} = 3.08$

e) The open economy multiplier is given by $1 / (1 - b + m)$. Therefore, the multiplier will be 1.8.
- 3) Go through Sub-Section 2.3.1 and answer.

Check Your Progress 3

- 1) (a) Go through Sub-Section 2.4.2; explain Fig. 2.7.

- (b) Saving will be lower than investment, since $(S + T) = (\bar{I} + G)$.
- 2) Refer to Fig. 2.8 and Fig. 2.9 for the answer.
- 3) (i) $S = -10 + 0.25Y$; (ii) 120; (iii) 112

Check Your Progress 4

- 1) You should discuss about transaction demand, precautionary demand and speculative demand for money.
- 2) Refer to Sub-Section 2.5.1 and answer.
- 3) It refers to the insensitiveness of the demand for money to rate of interest, when the rate of interest is very low. Conventional instruments of monetary policy become ineffective in liquidity trap.
- 4) Government should increase public expenditure so that there is no decrease in effective demand.
- 5) High fiscal deficit increases government debt burden, increases interest rate and may lead to crowding out of private expenditure.

UNIT 3 THE NEOCLASSICAL SYNTHESIS*

Structure

- 3.0 Objectives
- 3.1 Introduction
- 3.2 Equilibrium in the Real Sector
 - 3.2.1 Derivation of the IS Curve
 - 3.2.2 Shift in the IS Curve
- 3.3 Equilibrium in the Monetary Sector
 - 3.3.1 Derivation of the LM Curve
 - 3.3.2 Shift in the LM Curve
- 3.4 Simultaneous Equilibrium of Real and Monetary Sectors
 - 3.4.1 Shocks in the IS-LM Model
 - 3.4.2 Adjustment Process in the Economy
 - 3.4.3 Impact of Supply Shocks
- 3.5 AD-AS Model
 - 3.5.1 Derivation of the AD Curve
 - 3.5.2 Aggregate Supply Curve
- 3.6 Let Us Sum Up
- 3.7 Answers/Hints to Check Your Progress Exercises

3.0 OBJECTIVES

After going through this unit, you should be in a position to

- explain the equilibrium in the real sector and the monetary sector in an economy;
- explain the underlying ideas behind the IS curve;
- explain the underlying ideas behind the LM curve;
- explain how the IS and LM curves interact;
- identify factors that influence the position and slope of the IS and the LM curves;
- derive aggregate demand (AD) curve from the IS-LM model; and
- find out the factors that influence the AD curve.

3.1 INTRODUCTION

In Unit 2 we discussed how the equilibrium level of output is determined in the Keynesian model. Recall that Keynes came up with his analysis in the aftermath of the Great Depression. He considered an economy with underutilization of resources (i.e., presence of idle capacity) with an objective of correcting the

economy immediately. Thus, he focused on the short-run. He reasoned that an increase in aggregate expenditure (which represents aggregate demand) leads to an increase in output; it does not lead to rise in prices till full employment is reached. In view of above, price level is kept fixed in the simple Keynesian model. An implication of fixed prices is that nominal values and real values are the same.

In this Unit we deal with a closed economy; there is no external trade. We begin with the derivation of IS and LM curves. Subsequently, we derive the AD curve from the equilibrium points of the IS-LM model.

3.2 EQUILIBRIUM IN THE REAL SECTOR

The aggregate expenditure (or, aggregate demand) of an economy is given by

$$Y = C + I + G$$

where Y is output (or, aggregate supply), C is consumption expenditure, I is investment expenditure, and G is government expenditure.

The consumption function is given by

$$C = a + b(Y - T) = a + bY_d \quad \dots (3.1)$$

where Y_d is disposable income (i.e., $Y - T$). Here T is a lump sum tax levied by the government. The investment function is given by

$$I = c + dY - e i \quad \dots (3.2)$$

where i is the rate of interest (there is no difference between nominal rate of interest and real rate of interest as price is kept fixed). Let us assume that government expenditure (G) is constant, and exogenously given. Thus,

$$Y = a + b(Y - \bar{T}) + (c + dY - e i) + \bar{G} \quad \dots (3.3)$$

On re-arrangement of terms in equation (3.3), we obtain

$$Y - bY - dY = a - b\bar{T} + c - e i + \bar{G}$$

$$\text{Or, } Y = \frac{a+c+\bar{G}-b\bar{T}}{(1-b-d)} - \frac{e}{(1-b-d)} i \quad \dots (3.4)$$

The IS equation, in its general form, is given by

$$Y = \alpha_G(\bar{A} - e i) \quad \dots (3.4a)$$

where $\alpha_G = \frac{1}{(1-b-d)}$ and $\bar{A} = a + c + \bar{G} - b\bar{T}$.

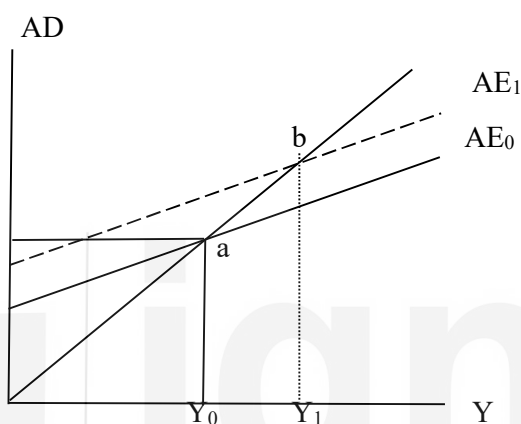
In equation (3.4) we find an inverse relationship between Y and i . It implies that, as i increases, there is a decrease in Y . In order to find out the slope of equation (3.4) you have to express it in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the IS curve as $-\frac{(1-b-d)}{e}$. Depending upon the value of the parameters, the IS curve would be flatter or steeper.

3.2.1 Derivation of the IS Curve

In Fig. 3.1 we explain the relationship between Y and i through a diagram. In Panel (a) of Fig. 3.1 we take aggregate expenditure (AE) on the y-axis and output (Y) on

the x-axis. Let us assume that the economy is operating with AE_0 while the rate of interest is i_0 . The intersection of AE_0 with the 45° line at point 'a' is of utmost importance to us. Note that the 45° line in Fig. 3.1(a) indicates equality between aggregate demand (AE) and aggregate supply (Y), such that it corresponds to equilibrium in the economy. We plot this point 'a' in panel (b) of Fig. 3.1. In panel (b) we take interest rate (i) on the y-axis and output (Y) on the x-axis.

Panel (a)



Panel (b)

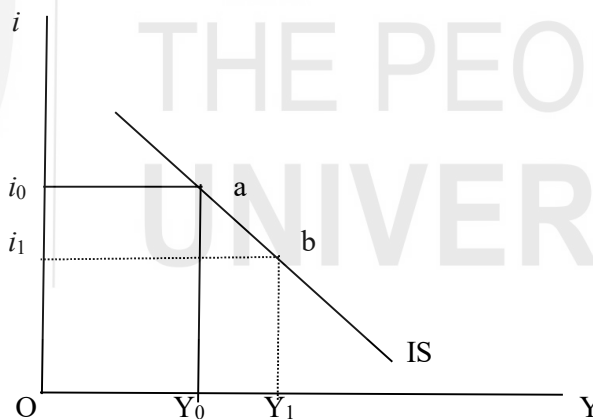


Fig. 3.1: Derivation of the IS Curve

Suppose, there is a decrease in the rate of interest from i_0 to i_1 . As a result of this, the level of investment in the economy increases and the new aggregate expenditure level rises to AE_1 . The aggregate expenditure curve (AE_1) intersects the 45° line at point-b. Due to the increase in the aggregate expenditure, there is an increase in the equilibrium level of output from Y_0 to Y_1 . We plot point-b in Fig. 3.1(b) corresponding to output level Y_1 and interest rate i_1 . When we combine points 'a' and 'b' we obtain a downward-sloping curve, called the IS curve. A point on the IS curve denotes equilibrium in the real sector (also called the goods sector) of the economy and there is equality between investment (I) and saving (S). Note that the real sector of the economy is at equilibrium on each and every point of the